

The use of focus groups in complex and pressurised IS studies and evaluation using Klein & Myers principles for interpretive research

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Abstract. *Studies of information systems (IS) often involve many complexities, uncertainties and restrictions. As a result, researchers have to adopt innovative research approaches or tailor and extend existing ones. This paper describes and evaluates an innovative focus group approach used to study three IS development teams, which had to cope with the application of a broad, polymorphous framework, untested in practice, seeking to elicit potentially highly sensitive opinions and judgements in a highly pressurised, time-restricted environment. This paper makes two significant contributions. To conduct this evaluation, the researchers operationalised the Klein & Myers seven principles for interpretive study for the first time, creating a set of questions that other IS researchers can now use to evaluate their own interpretive studies. The paper also applies these to the design of the focus groups and develops a set of recommendations for future IS researchers to consider based on lessons learned in this study.*

Keywords: applied focus group approach, Klein & Myers, interpretive research evaluation, hermeneutic circle, agility assessment

INTRODUCTION

The focus group is an underutilised research approach in the information systems (IS) discipline despite being seen as very suitable for IS studies. This is partly because of the fact that IS studies are often complex, pressurised and involve uncertainty and restrictions. As a result, researchers must tailor and extend the focus group approach to cope with these issues. In the IS study presented in this paper, for example significant time restrictions were in place, a previously untested framework was used to structure the study, the

framework was broad and multifaceted, the underlying constructs were highly complex and multidimensional and the original framework and sub-concepts had no previous accompanying set of prescriptive questions. The study involved geographically distributed team members, there was significant potential for poor judgement, unsupported anecdotal statements, statements arising from ulterior motives, and the data were highly sensitive and controversial. All of these issues required careful consideration when designing the focus group approach for the study.

The quality of research using such an adapted focus group approach must be evaluated rigorously. This is especially true where the study is of an interpretive nature and where evaluation approaches are traditionally based on positivist concepts of validity and reliability. As the popularity of interpretive research in IS has increased, a debate has developed as to how to evaluate its quality. The Klein & Myers (1999) (K&M) principles have been proposed as a solution, but a review of the literature reveals that while the principles are often cited, they have not been operationalised, thus limiting their application. Although different approaches to operationalisation of the principles are possible, here we adopt what we consider an approach widely applicable to interpretive research, it being based on the concepts underlying the principles themselves rather than any particular research method. In this paper, we explain how the operationalised framework was developed, and we demonstrate its application through the evaluation of a focus group study.

Thus, the objectives of this paper are as follows: (i) to outline the focus group approach used and the researchers' rationale underpinning the decisions behind the focus group design; (ii) to operationalise the Klein & Myers (1999) principles; (iii) to evaluate the quality of the focus group approach using the operationalised principles; and (iv) to propose a set of recommendations for future applications and tailoring of this approach. To this end, the next section of this paper introduces the focus group approach and reviews the application and evaluation of the focus group technique in IS development (ISD) research. We then provide an overview of the literature relating to the evaluation of interpretive research. In the same section, we propose and operationalise the seven principles to be followed by interpretive research (Klein & Myers, 1999) as a set of self-reflective questions. The following section describes the focus group approach adopted in this study. Subsequently, we evaluate the applied focus group approach according to those criteria and distil a set of lessons learned from this experience and recommendations for future researchers contemplating the application of our focus group approach to their research. The paper concludes with the contributions of the study and possibilities for future research.

FOCUS GROUP APPROACH AND EVALUATION

The focus group approach is defined as a 'research technique that collects data through group interaction on a topic determined by the researcher' (Morgan, 1997) and involves a group of participants and one or more moderators. The core theoretical elements of focus groups include topical focus, group interactions, in-depth data and a 'humanistic' character

(Stewart *et al.*, 2007). The focus element derives from participants of the group having a 'particular concrete situation' in common (Merton & Kendall, 1946) providing a basis for effective communication. The researchers guide discussions with suggested topics and probing, open-ended, questions. However, the hallmark of focus groups is the group interaction through which insights and less accessible data can emerge, which may not otherwise come to the surface. Researchers (Kitzinger, 1994; Kidd & Parshall, 2000) draw attention to the importance of this differentiator of focus groups from other forms of collective or focused interviews – that is group interaction and discussion. This aspect is particularly important where participants may not know much about the research topic or require a group discussion to stimulate them to contribute (Morgan, 1997). Merton & Kendall (1946) refer to this as 'introspective retrospection'. In some ways, focus groups could be considered to lie between dyadic interviews and direct observation: while allowing the researcher to direct attention to specific topics as allowed by interviews, they also facilitate group discussion as per observation (Kitzinger, 1994). As well as the richness of data collected, another significant benefit of focus groups is the ability to get many data from a group in a short amount of time (Morgan, 1997; Stewart *et al.*, 2007). In addition, focus groups are suitable for 'formative evaluation for programme improvement' (Patton, 1990), which matches the use in this study (Hines, 2000).

All participants in the project we studied shared common membership of an ISD project team. As the topic being investigated related to the team as well as individuals, the group aspect of the technique is considered important in increasing the depth of data collected, as discussions stimulated reflection and helped surface opinions and other inputs that might otherwise not have been forthcoming. Given the topic guide was based on the conceptual framework of agility and that this was entirely new to the participants, the authors suggest that the group environment also helped arrive quickly at a common understanding of the framework. Using the agile concepts as a lens to consider past experiences led to lively group discussions, joint exploration of new ideas and rich and plentiful data. It also contributed to a joint sense of ownership of the emerging findings and potential recommendations from the research.

Focus groups in IS research

There are many reasons for specifically evaluating focus group research in the IS discipline. First, despite being extensively used in other disciplines such as marketing and health service research, an examination of the IS literature shows that focus groups are valuable but underutilised, with very few IS studies adopting the approach to date (Sobreperez, 2008) and none specifically within the domain of ISD as far as we are aware. This is despite the fact that focus groups seem very suitable to IS research – many advantages discussed earlier have clear relevance to the IS discipline, such as overcoming limited access to data and suitability to unexplored and emerging topics. For example Chesney *et al.* (2009) use online focus groups to explore grieving behaviours in virtual worlds. Of the few instances of focus group adoption in IS, most relate to the application of IS in contexts where focus group studies are more widely used, such as health (Eysenbach, 2000; Koppel *et al.*, 2005) and marketing (Pitt

et al., 1995). Some other research in IS has used focus groups but only as a minor supplementary data collection technique (Reich & Benbasat, 2000). More indirectly, related fields such as library and information science have reportedly used focus groups extensively (Kerlake & Goulding, 1996).

Another reason for critically evaluating the use of focus groups in IS is the extent to which we handle the *potential drawbacks associated with the approach*. Group interaction can lead to effects such as conformity of views as dominant characters or roles in a group causing others to 'tow the line' (Morgan, 1997). Conversely, it can lead to polarisation into one or more subgroups or 'factions' (Sim, 1998; Barbour, 2007), and thus opinions expressed may reflect those of the group rather than the individuals. Small group dynamics come into play, which can stymie minority or controversial opinions. Opinions expressed may reflect those of a particular group context rather than an aggregation of the opinions of the individuals (Stewart *et al.*, 2007). Additionally, more 'subtle' input that might emerge in a less 'public' context may be missed. While the moderator has less control over a group and less access to individual opinions, the setting of the topic of focus by the researcher can also influence the range and depth of input gathered and may not reflect topics considered important or interesting to participants (Merton & Kendall, 1946). Focus groups do not allow observation of groups in more 'natural' contexts and are primarily restricted to discussions rather than other forms of interaction (Morgan, 1997). Data collected are confined to participants' opinions and reportage of experiences and therefore are highly subjective (Wilkinson, 1998).

Next, we should critically reflect on the use of focus groups as, while the literature contains some general 'rules of thumb' (Morgan, 1997), the execution of focus groups remains quite ambiguous and ill-defined compared with other techniques such as surveys where best practices are relatively well known and mature (Kidd & Parshall, 2000). For example Morgan (1997) suggests that when conducting focus groups, the researcher 'should use a relatively structured format with high-moderator input'. This in itself is quite a loose prescription, but the true ambiguity is revealed when Morgan suggests that 'this guideline will vary depending on how exploratory, descriptive or explanatory the research objective'.

Finally, although a few texts broadly describe how to design effective focus group research, there are few criteria defined for their evaluation. Previous studies have shown that focus group researchers need to consider issues around consensus and dissent, strength of opinion and generalisation (Sim, 1998), and range, specificity, depth and personal context (Merton *et al.*, 1990). However, from our experiences in this study, there are many other threats to the quality of the research that need to be taken into consideration when conducting focus group research. Therefore, the authors turn to more generally applicable evaluation frameworks that consider the whole research design and execution rather than confining themselves to the focus group data collection method specifically.

EVALUATING INTERPRETIVE RESEARCH

As discussed in the introduction to the paper, interpretive research in IS has gained considerably in popularity since 1991 when it constituted a very small proportion of published IS

research (Orlikowski & Baroudi, 1991; Walsham, 2006). Early interpretive research used quality criteria borrowed from the well-established positivist tradition that broadly assesses validity and reliability at various levels of granularity and specificity. Texts on qualitative and case study research offered operationalised versions of these (Miles & Huberman, 1994; Yin, 2003) and were adopted by interpretive researchers (Johnson *et al.*, 2006). However, there has been an increasingly active debate in the literature over the appropriateness of such criteria, as they are in positivist philosophical assumptions (Johnson *et al.*, 2006). Recognition of this weakness has led to three distinct responses in the literature. One contends that the epistemological underpinnings of interpretive research argue against the use of established, standardised or imposed criteria at all (Bochner, 2000), which is seen as constraining the freedom and limiting possibilities of the research – ‘personal standards’ should instead ensure the quality of the research. However, as an objective of this paper is to provide guidance to other researchers using focus groups in IS research, this stance is rejected as not contributing to our present goal. A second response is that of extending existing criteria for validity and reliability by redefinition (Lincoln, 1995) and subcategorisation (Kirk & Miller, 1986; Altheide & Johnson, 1994). However, these approaches remain embedded in the underlying positivist philosophy, continue to be hampered by the resulting ‘epistemological quagmire’ (Whittemore *et al.*, 2001) and have even been associated with undermining research rigor (Morse *et al.*, 2002). A third response is to propose alternates that are not based explicitly on validity and reliability. Two of these are proposed for IS research (Walsham, 2006) and were considered for this study.

Golden-Biddle & Locke (1993) propose the use of authenticity, plausibility and criticality in assessing the ability of ethnographic research texts to ‘convince’ the reader. They identify a range of devices and techniques used by authors in preparing such texts to demonstrate the rigor of their research to their audience. However, much of their argument is specific to ethnography and is an example of a post hoc evaluation of reportage of results rather than actively contributing to the overall quality of research (Morse *et al.*, 2002). An alternate and ‘more comprehensive’ (Walsham, 2006) framework, developed specifically for the *conduct and evaluation* of interpretive research in IS, is the set of seven principles based on anthropology, phenomenology and hermeneutics that have gained prominence in recent years (Klein & Myers, 1999) – simply referred to as K&M in the remainder of this paper. K&M are frequently cited (approximately 1800 citations as of December 2010) by IS researchers to justify or validate their research approaches (e.g. Pan *et al.*, 2005; Dhillon & Torkzadeh, 2006; Bjørn & Ngwenyama, 2008).

There are seven K&M principles, the first being an overarching, fundamental meta-principle of the ‘hermeneutic circle’ that applies to the remaining six principles of *contextualisation*, *interaction*, *abstraction and generalisation*, *dialogical reasoning*, *multiple interpretations* and *suspicion*. All are briefly described in later sections of this paper. These principles have been recognised as a ‘critically important. . . set of criteria on how to judge knowledge claims generated from interpretivist research’ (Hirscheim & Klein, 2006). They have been cited as particularly important in terms of establishing the generalisability of interpretive research (Lee & Baskerville, 2003; Gregor, 2006). There are also instances of their reference in developing

criteria for more specific research methods such as canonical action research (Davison *et al.*, 2004) and design science research (Hevner *et al.*, 2004). However, a search of the literature has not revealed any previous studies involving the formal operationalisation of the principles. In developing a set of criteria that operationalise them, the researchers performed a search for key words and phrases in the descriptive text describing each principle in the paper where they were first elaborated (Klein & Myers, 1999). Where such key text is judged to represent significant and distinct concepts and constructs of the principle, these are used to guide the development of one or more questions to evaluate any interpretive research method with respect to that principle. The matching of key text and the resultant questions are presented in Table 1.

THE FOCUS GROUP APPROACH USED IN THE PENNYSOFT CASE

This study was tasked with assessing the agility of a proprietary method in a globally distributed systems development organisation, part of an emerging empirical research area relating to adoption and assimilation of methods, such as eXtreme Programming (XP) and Scrum, in practice. Rather than assessing agility by the compliance of implementation with the documented method, we examined three teams using a conceptually well-established agile assessment framework (see Conboy, 2009). This framework identifies the core underlying attributes of agility, allowing any practice to be assessed from first principles. The assessment objectives were (i) to establish the agility afforded by the proprietary method as implemented in the organisation ('method in action') and (ii) to identify major issues with the adoption and generate recommendations to address them. Three major data collection phases were involved, namely, document analysis, semi-structured interviews and, finally, in-depth focus groups. While the first two phases were relatively straightforward, the focus group phase and the environment within which it was conducted posed a number of significant challenges.

This study was highly complex for a number of reasons, driving the need for an innovative focus group approach. First, because of other severe product deadline pressures, *significant time restrictions* were in place with the research personnel limited to a 3-hour window with each ISD team under scrutiny. Secondly, a *previously untested framework* was used to structure the study, meaning the researchers were operating largely in 'unknown territory'. In addition, the framework was *broad and multifaceted*, comprising of a range of diverse concepts, namely team creativity, proaction, reaction, learning, simplicity, economy and quality. Each of these *constructs was highly complex and multidimensional* in their own right, increasing the chance of misinterpretation or multiple interpretations during data collection. Many have highlighted the complex and multifaceted natures of creativity (e.g. Henry, 1991; Woodman *et al.*, 1993; Amabile, 1996), learning (e.g. Huber, 1996; Argyris, 1999) and quality (Shiba *et al.*, 1992). Given the complexity and broadness of the parent framework and its sub-concepts and the lack of an *accompanying set of prescriptive questions*, careful design of the focus groups was required to ensure that data of sufficient quantity and richness was captured across all the agile concepts while restricting tangential, irrelevant data. The study was also based on eliciting the

Table 1. The evaluation framework of interpretive study (based on K&M)

K&M principles	Self-reflective questions derived by the authors from K&M keywords	K&M keywords
1. The fundamental principle of the hermeneutic circle	Did the research method support looking at the parts, the whole and relationships between these in developing a comprehensive understanding?	Whole, parts, relationships
	Did the research method support an evolving understanding by facilitating changing preconceptions of the parts, the whole and their relationships as the study progressed?	Movement of understanding, preconceptions about the meanings
	Did the research method allow a 'complex whole of shared meanings' of the parts, the whole and their relationships across the researchers and participants (where appropriate) to emerge?	Shared
	Did the research method support reconciliation of contradictions or gaps in understandings, leading to a harmonious overall understanding?	Harmony, fill in any gaps or unresolved contradictions
2. The principle of contextualisation	How did the research method help identify and elaborate the historical, social, political and economic contexts of the concrete situation and aid in the understanding of how this would affect the research? (Examples of context include roles, seniority, experience, skills, corporate culture, vision and business environment)	Historical, social, political and economic contexts
	How did the research method capture the moving and evolving nature of these contexts and aid understanding of how this would affect the research?	Moving target
	How did the research method affect the ability of the research to reflect the various contexts in the analysis and reporting, depending on the audience and purpose of the study, with the aim of bringing out how the current situation emerged?	Explicitly reflect, bring it out
	Did the research method capture how the history of the participants, the researchers and the various contexts both influenced and contributed to, as well as produced, the present, concrete situation under study?	Subjects are agents, producers of context
3. The principle of interaction between the researchers and the subjects	How did the research method, such as group discussions and written input, reveal how the data were co-created through interaction of the researchers and subjects during data collection and interpretation?	Data collection and interpretation affected one another
	Did the research method reveal explicitly the researchers/participant historical and social relationships and explore how such social construction may affect the data?	Facts are produced as part and parcel of the social interaction, socially constructed
	Did the research method reveal how the value-laden, subjective interpretations of the participants affected the data collected, and how such interpretations were themselves affected by the research method?	Participants can be seen as researchers and analysts
	Were the data supplied by the participants constrained or otherwise influenced by the researchers' preconceptions?	Researchers' preconceptions
4. The principle of abstraction and generalisation	How did the research method help researchers and participants become sensitive to significant or important data – i.e. those that contributed to concepts, theory, insights or implications?	Sensitising device
	How did the research method help relate collected data to abstract categories or to identify new concepts and relationships between them?	Abstraction, relate particulars to abstract categories
	How did the research method help justify this categorisation – did it help explain how the concepts and constructs were plausible and logical?	Plausibility and cogency of the logical reasoning
	How did the research method support the relating of idiosyncratic data from specific participants or concrete situations to generalisations, which may apply across multiple situations, ensuring that readers can follow how the research team arrived at their theoretical insights?	Generalisation related to study details [. . .] so readers can follow

Table 1. cont.

K&M principles	Self-reflective questions derived by the authors from K&M keywords	K&M keywords
5. The principle of dialogical reasoning	How did the research method help the researchers become aware of their own historicity and prejudices, and how that affected the research design and conduct even though it serves as a necessary starting point, the original lenses, of their understanding and cannot be simply set aside?	Our own historicity, the original lenses
	How did the research method facilitate or aid the identification of true from false prejudices and the changing of such preconceptions over time to avoid contradictions in the light of emerging data?	Distinguish true prejudices from false ones, sensitivity to possible contradictions
	How did the research method help make the evolution of prejudices and preconceptions throughout the research as transparent as possible, thereby minimising the adverse effects of researchers 'false' preconceptions?	Transparent
6. The principle of multiple interpretations	How did the research method allow expression, exploration and explanation of multiple viewpoints of participants of the same sequence of events, for example a diversity of viewpoints within the team such as mental models related to power, economics or values (if they existed)?	Possible sensitivity to differences in interpretations, multiple viewpoints, same sequence of events
	Did the research method help surface conflicts and contradictions resulting from such differing viewpoints?	Conflict
	Did research method allow confrontation of these conflicting viewpoints and address their effect on the concrete situation and the research?	Confrontation
7. The principle of suspicion	Did the research method allow critical probing and seeking of justifications of the teams' interpretations in terms of possible (conscious or unconscious) distortions caused by factors such as power structures, vested interests and limited resources?	False consciousness, critical
	Did it help identify the effects of dominant individuals or viewpoints (e.g. 'politically correct' statements), a lack of 'openness' or 'truthfulness' between team members and other factors affecting the social construction of reality that can lead to distortions of the data collected?	Domination, asymmetry and distorted communication, distortion in narratives
	Did it facilitate questioning of participants 'reality' to determine influences in the social construction of that viewpoint, such as alignment with dominant organisational perspectives, leading to alternate viewpoints being obscured or misrecognised?	Obscured and misrecognised

K&M, Klein & Myers.

subjective opinions of the ISD team members and the approach therefore had to address the *potential for poor judgement, unsupported anecdotal statements and statements arising from ulterior motives*. The self-appraising nature of the study also meant that *sensitive and controversial data* could arise, such as criticism of peers or project managers, and so careful attention had to be paid to ensuring this was nurtured and captured, but done so in a sensitive manner. Also, some projects involved *geographically distributed team members* presenting additional issues in terms of data collection.

Pennysoft¹ is a global financial services firm with approximately 45 000 employees world-wide and up to 10 000 of these developing systems to support the business, distributed across multiple sites in the US, Europe and India. A formal 'waterfall' development method

¹Pennysoft is a pseudonym used for the purposes of confidentiality.

has been used extensively across the company for several years. A newly developed proprietary method incorporating many principles and practices from agile methods such as Scrum was being piloted at the time of this study. Three such trial projects located in an Irish office were studied between July and September 2009. All were part of distributed teams, but with the majority of analysis, development and test based in Ireland. One was a 'green field' project with some US members and a small, inexperienced team of five. The other two were larger (10–20) with US- and India-based members and were part of larger enterprise-wide programmes.

In preparation for the focus groups, researchers first analysed the method documentation and training material to establish the documented method, which was common across the projects. This understanding was used to develop an interview guide for semi-structured dyadic interviews with project managers from each team which, along with observation of key team practices, provided an understanding of the method as it was implemented in each project, the method in action (Fitzgerald *et al.*, 2002). Significant differences were found across the projects, with different sets of practices implemented in different ways.

A single, 3-hour focus group was then held for each project, with a representative sample of half or more of the team selected by management. In one case, two participants based in India used a videoconference link and contributed written input using instant messaging. Between three and six researchers filled the roles of moderators, note takers and discussion facilitators, and several audio recorders, pens and post-its were placed about the room. Following an overview of the research objectives and the format of the session, the agile conceptual framework was introduced and displayed on wall posters for the team's reference, including examples of how a practice might impinge on the underlying agile concepts. Then, 5–10 minutes were spent considering each agile practice in turn using pre-prepared A1 wall posters briefly describing the practice in bullet form as used in that project. The moderator described briefly each practice in turn to refresh participants' memories, inviting corrections where necessary and encouraging discussions and written input. Each poster included a column for each agile concept (see Figure 1) where participants could stick post-it notes (which were anonymous) with their input on how that concept was affected by the practice concerned, effectively associating each post-it with an agile practice and concept. Periodically, by highlighting vague or abstract contributions from the group, the importance of specific, concrete examples was emphasised by the moderator. This was followed by 30 minutes for further discussion of emerging topics, ambiguities and so on. At the end of each session, participants were invited to submit any further, possibly confidential, input directly to the researchers outside the focus group setting.

Immediately following each focus group, the post-it notes were coded to record the practice and concept to which they applied and the content was typed up into data tables for use in the analysis. Group discussions, which took place during the session, were transcribed from the audio recordings. The initial analysis was allocated to four researchers with each working on a set of practices, sometimes jointly. Because of time and resource constraints, formal coding and other data analysis techniques were not used. Following the analysis, each section was reviewed by at least two academic reviewers and one industrial reviewer (all of whom had also

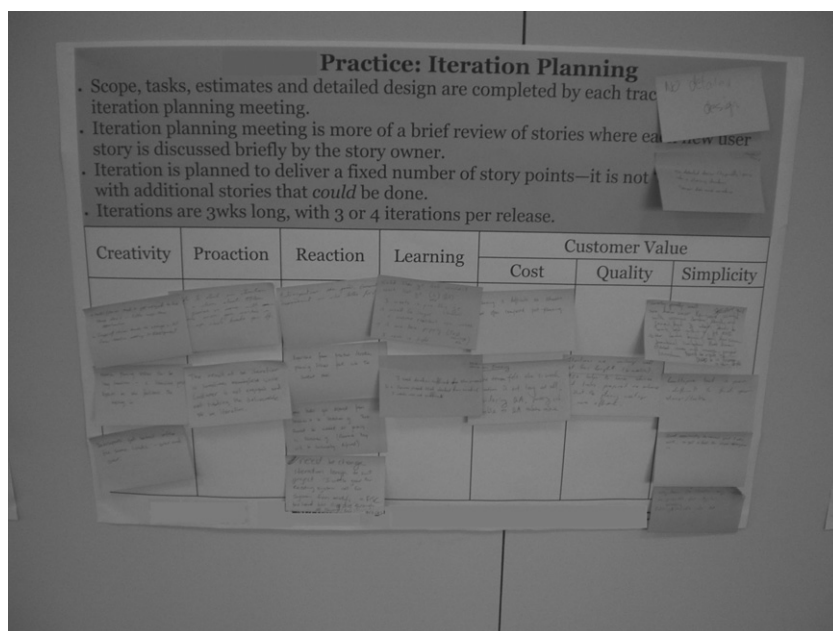


Figure 1. Poster sample with post-it note contributions.

been involved in the focus groups) with extensive discussions and revisions leading to a comprehensive report for the company.

The distinctive aspect of the research method lies in the various modifications and extensions used in the focus groups. Introduced primarily for efficiency of data collection, the evaluation questions will focus on how they may otherwise affect the research method positively or negatively. Practices such as written participant input, concurrent group discussions and a highly structured format may have unexpected effects on the plausibility and authenticity of the research. Such effects may originate in the focus group data collection but have consequences extending to research design, analysis and reporting. Thus, while evaluation concentrates on the applied focus groups, other elements of the overall method are also considered during the evaluation.

EVALUATION OF THE APPLIED FOCUS GROUP APPROACH

The fundamental principle of the hermeneutic circle

The first of the seven K&M principles, the hermeneutic circle, is regarded as a 'meta-principle' and applies to all the other six principles. As this principle calls for iterative consideration of both the *parts and whole* of that under study, as well as the *relationships* between them, it is

appropriate that what constitutes these parts, whole and relationships are discussed. The interpretation should be 'broad and liberal' and may evolve both as understanding develops and as different hermeneutic concepts are addressed (R1)². In terms of this study, therefore the parts and whole included respectively the individual method practices and the agility of the method, the seven constituent agility concepts and the team agility to which they give rise, and the individual participants and the team they form.

Aligned with this principle, data collection occurred through three distinct *iterations*: document analysis, dyadic interview and focus groups. The details of each stage were informed by the understanding gained during preceding data collection and analysis, leading to highly structured focus groups designed specifically for each ISD team (R2). Within each of these stages and specifically in analysis of the focus group data, further iterations of analysis of both written and verbal data, followed by peer review from other researchers, brought further refinements to understanding of the whole (R3).

The theoretical basis of the study was the disaggregation of the ISD method (whole) into its constituent practices (parts) and agility (whole) into its various concepts (parts). The focus group approach supported looking at each of these parts, although the relationships of these to the whole were derived by the researchers from the underlying theory. Participants were not asked to consider how individual practices contributed to the agile method or how agile concepts contributed to agility. Also, participants did not consider relationships between such parts. Where the research setting allows, further exploration of such relationships by participants may lead to deeper insights and contributions to theory (R4). More generally, although the researchers concluded that they had reached a *harmonious overall understanding* of the whole by the end of the study, more extensive dissemination of interim and final findings to participants and consideration of their feedback may have exposed further contradictions or gaps in knowledge (R5).

The following sections explore aspects of the hermeneutic circle more comprehensively by examining the study from the perspective of the six specific principles.

The principle of contextualisation

Explicit reflection of the context of both researchers and participants and the possible influence of those can significantly improve the rigor of interpretive research. Prior to the focus groups, the researchers explored the contexts of teams, projects and the organisation using document reviews and dyadic interviews. Through an understanding of the defined, documented method and the method in action, a deeper understanding of the context was gained (R6). The method-in-action framework proposes that the implemented method is influenced by both 'overt' or 'rational' roles and 'covert' or 'political' ones. It is argued here that by guiding the interviews and analysis to cover both these factors, the varied *contexts such as political, social, historical and economic* proposed by this principle are addressed. Interpretation of both written and interview data can be significantly influenced by the subjectivity of both the interviewee

²Each recommendation is assigned a label (R1, R2, . . . Rn). All recommendations are then listed in Table 2.

and researcher. Therefore, interviews by multiple researchers with multiple team members and analysis of documentation by more than one researcher are recommended to ensure that an accurate and complete understanding of the method in action emerges (R7). In the present case, some inaccuracies in our understanding of the method in action, because of differences of opinion, perspective or recollection of participants, were discovered during focus groups. Involving more researchers and participants in the document reviews and interviews, combined with direct observation of practices and validation of findings with participants, is also recommended to help triangulate the data and isolate such invalid interpretations (R8).

Although the documented method used by a team may be stable for periods of years, their implementations are *constantly changing* because of complex contextual factors and experiences such as shifting business priorities, technological change and new learning. Therefore, it is recommended that sources of contextual change are identified early in the research and steps taken to minimise their effects (R9), such as running the focus group shortly after establishing context, identifying and compensating for *moving targets* and *explicitly reflecting* on the effects of these changes on the research. Also, explicit description of such context in reports is an important factor in ensuring transparency of the research. By bringing out the context that gave rise to the focus group structure, the researcher improves the quality of the focus group research itself (R10).

As the method in action was implemented by the team itself based on the documented method, participants can be expected to have acted as *agents* in producing the context of the study, as well as being influenced by it. More in-depth data collection and analysis across multiple team members may have surfaced how team members acted as such agents and are not only passive subjects of the context (R11). Researchers should consider carefully the effect that both participants and researchers can have on the concrete situation under study. For example participants may have invested effort in certain aspects of the method, project or team, and therefore, their data input on those may not be entirely objective. Similarly, researchers may have special interest or knowledge in certain aspects of the concrete situation leading to uneven or biased treatment in data collection, analysis and reporting. Such investments or influences should be *explicitly stated* and considered by the researchers, with likely effects described in reporting the research (R12).

The principle of interaction between the researchers and the subjects

Interactions between researchers and participants can influence interpretive research. In this case, descriptions and illustrations of the agile concepts used, the definitions of method practices and the order and emphasis placed on topics during the focus groups may have influenced participants. For example addressing a certain practice early in the session may result in more attention and energy from the group being directed to it. The careful use of open-ended questions by the researcher and avoiding offering opinions or judgements during interview (R13) may help reduce the impact of these effects on the research. Similarly, the emphasis, depth of knowledge or interest of participants in certain topics may have affected

researchers' views. *Recording and reflecting* on such possibilities can help compensate and maintain a more balanced interpretation of the data (R14).

In the present case, multiple researchers were involved in conducting the focus groups. Given the time constraints, number of participants and wide variety of topics to be covered, it was found that defining distinct roles such as facilitation, recording of group discussions and monitoring written input, and then assigning these to different researchers was necessary. It is also noted that a variety of skills among the researchers such as facilitation, domain knowledge and industry experience was helpful in running the sessions efficiently (R15).

Because of time constraints, it was not possible to validate the data collected or the research findings with participants after the focus groups. This restricted the opportunity to identify and compensate for *subjective interpretations of the researchers or, indeed, the participants* in the data they provided. Anecdotally, some company managers who had not participated in the research disagreed with some of the written input provided, especially where it described shortcomings in the method or organisation. Even though none of the researchers' findings were called into question, this disagreement may have undermined the perceived validity of the research. The anonymous nature of the post-its may have facilitated participants in providing politically motivated input. Post-focus group validation of input with the participants (R16) is considered important in tempering this tendency and provided an opportunity to contest any controversial statements before they unduly influenced the research findings and credibility.

Although the focus groups were structured around the method-in-action and agility concepts for each project, their conduct allowed more time and attention to those practices, attracting more interest and input from participants. Conversely, where there was little interest in a practice, the time spend discussing it was reduced. This limited flexibility was in addition to participants correcting or adding to the researchers' descriptions of each practice before discussions began or adding new practices. Therefore, although the overall structure of the sessions was dictated by interactions between researchers and participants and was designed for efficient data collection, some latitude was allowed for participants to discuss unforeseen topics where appropriate (R17). Also, participants should be invited to provide additional input after the focus group is over, where time constraints, sensitive issues or data not directly related to the topics are concerned.

The principle of abstraction and generalisation

This principle is summarised by K&M as 'relating the idiographic details revealed by the data interpretation through the application [. . .] to theoretical, general concepts that describe the nature of human understanding and social action'. The design and execution of the focus groups were *informed by theory* (R18), with the sub-headings on the wall posters reflecting the ISD method-in-action and the agility concepts. The approach allowed participants to instantly *relate* the data by associating their input (post-it note) with both the agile practice and the agility concept to which it referred. Although the practices are unique to the method enacted by a particular project or team, the agile concepts are abstract and assigning data directly to them allowed the participants *self-code* their written input to abstract categories. This increased the

efficiency of data collection and analysis (by sensitising the researchers and their instruments (R19)) while also improving the relevance of data to the research (by sensitising the participants (R20)). Data derived from group discussions in the focus groups were also assigned to these categories during data analysis, a task facilitated by the theoretical frameworks used to frame such discussions. These benefits rely on a shared understanding by the researchers of the theories and their operationalisation in the focus group.

Beyond duplicating post-it notes, the wall posters did not allow input to be associated with multiple concepts or multiple practices. Such a restriction could have been addressed using a mechanism to allow more flexible associations between written input, agile concepts and practices (R21). This may have allowed new concepts or constructs emerge from participant coding of input and is recommended for future research efforts. Where participants illustrate their opinions with examples or cogent arguments (R22), it was found that the nature of the relationship between concepts became clearer, greatly enhancing the researchers' ability to further refine abstract concepts and constructs.

In line with the interpretive tradition, generalisability in this principle is of a theoretical rather than statistical nature. Walsham (1995) identifies four types of generalisation in interpretive studies: the development of concepts, the generation of theory, the drawing of specific implications and the contribution of rich insights. The latter two are considered more relevant to the purpose of this study with its highly contextualised focus group approach. *Specific implications* that may be 'valuable in the future in other organisations and contexts' can be drawn relating to the manner in which agility is supported and inhibited by specific use of certain practices. However, such implications not only depend on robustness, on a thorough understanding of the team, project and technology context (R23) in which they were observed, but also transcend such context for wider applicability in the future. The development of detailed contextual descriptions of each of the three cases under study using the method-in-action framework can aid the plausible and logical analysis of the cases, allowing the consideration of the idiosyncratic nature of each and its effects on the findings. *Rich insights* can also be gleaned from a deep understanding of the case contexts and how that context impinges on multiple practices and their effects on agility, revealing tendencies that may apply to other contexts. Also, the careful selection of cases and participants to reduce variability in context (R24) along with attribution of input to individuals and their roles (e.g. by pre-coding post-its to individuals) (R25) is recommended to enhance generalisability.

The principle of dialogical reasoning

Interpretive research recognises that the researcher cannot cast aside all preconceptions and that understanding evolves with emerging empirical data. In this case, an understanding of the discrete practices in use and the agility concepts served as a *conceptual starting point*. Each practice was described concisely on the wall posters and reviewed by the focus group, with participants invited to *critique* the researcher's understandings of how it was implemented on the project (R26). In some cases, inaccuracies were found, which were because of the

practices evolving since the time the method-in-action analysis was performed and occasional differences in recollection and opinions between team members.

The *transparency* of the researchers' preconceptions afforded by the wall posters describing practices and agile concepts also captured the evolution of such preconceptions throughout the study. *Corrections* and comments were invited from participants and noted on the posters before each topic was discussed (R27). Preparation of the practice descriptions from documentation and dyadic interviews was also subject to peer review within the research team, while analysis of focus group data was conducted by pairs of researchers before wider peer review (R28). As written input was focused on the discrete set of agile concepts, differing understandings and ambiguities about these surfaced early in the data collection stage as participants tried to relate their input to specific concepts. After the first focus group, the language used to describe one concept was modified from 'economy' to 'cost' to increase clarity and delineation, and this encouraged a more uniform understanding and interpretation across participants (R29). During analysis, the data relating to each practice/agile concept pair were assessed in aggregate as to whether they supported agility, inhibited it, expressed conflicting opinions or indicated no strong opinion either way. The results of this analysis were tabulated and colour coded. This individual analysis of each part of the method in relation to each agile concept (R30) allowed effective comparisons of the method used in each project while maintaining transparency to the underlying data used to construct the aggregate view of the 'whole'.

In supporting the identification and abandonment of *false preconceptions* through sensitivity to possible *contradictions*, the research used three iterations of data collection (document analysis, method in action and focus groups) and analysis for each case. However, as each of the three steps was completed for all cases concurrently, lessons from one case could not be reflected in subsequent cases. Therefore, the theoretical frameworks used by the researchers in developing this understanding were not critically re-evaluated throughout the study. By completing the study of one case, *reflecting critically* on the research experience and results and reassessing the theories and *preconceptions* of the researchers, more thorough dialogical reasoning may have improved the quality of the research (R31).

The principle of multiple interpretations

The same sequence of events can be interpreted differently by multiple participants. Discussions between participants in focus groups can be influenced by small group dynamics, leading to either nominal consensus of opinion or alternatively polarisation along factional lines. The use of substantially anonymous written input in this study allowed *multiple, often divergent, views* on specific topics to be expressed (R32). Although participant discussions may have tempered the diversity of views that might result from other data collection approaches such as nominal group technique, the results of analysis show that in 10% of cases, conflicting opinions on the same topic were expressed. The researchers did not seek out or explicitly encourage diverse viewpoints during the focus groups. Although gaps in contributions for particular practices and agile concepts were actively addressed during the sessions by encouraging

further input, where there were multiple post-its submitted on a topic, little regard was given to the level of diversity of such data. To ensure that multiple viewpoints are captured, soliciting of divergent opinions could be actively promoted, thereby helping surface conflicts and contradictions.

Although some of the variability of opinion may be because of the ambiguous nature of the agile concepts, the research did not investigate its cause. An opportunity for participants to reflect on specific, jointly experienced events may have illuminated the underlying reasons. Although such concrete situations were sought from participants to illustrate their opinions, alternate views to explore further such events were not sought from others in the group. Participants were therefore not afforded the opportunity to *confront* the multiple views that surfaced during the research and to explore the *conflicts* and contradictions that might result from them. It is recommended that future studies consider such reflective consideration of shared concrete experiences (R33), perhaps after initial analysis of the focus group data. These might include *individual interviews where the potential conflicts and contradictions are of a sensitive nature*. Additionally, attribution of input to individuals and roles (R34), as recommended elsewhere in this paper, may aid understanding the causes of diverse opinions, such as power relations, economic considerations or social effects.

The principle of suspicion

A *critical* approach to participant input was adopted in the study to address possible biases and systematic distortions in the focus groups. Researchers carefully monitored group discussions and facilitated the inclusion of all team members rather than allowing certain participants *distort* discussions. This was aided by the antecedent analysis of the method in action for each project that included consideration of the team and project context, including covert, political aspects. A large proportion of written notes were visually checked during the sessions, and where concrete examples or other justifications for opinions were not included, participants were reminded of the need to support their input with low-inference descriptors (R35). However, analysis did not extend to a detailed assessment of power relations, economic priorities and other factors that could lead to *conscious or subconscious distortions*. In future studies, it is recommended that steps be taken to further surface such underlying influences in participants. These may include further analysis of participants and their power and economic priorities, and further probing of distorting influences through examination of data collected and follow-up dyadic interviews. In addition, triangulation between written input and group discussions may help discover distortions of data.

At the beginning of each focus group, participants were encouraged to provide truthful, unguarded input. Anonymity was assured and use of post-its was proposed where sensitive data were involved. On reflection, the researchers conclude that these measures served as a 'double-edged sword': while successfully eliciting sensitive and potentially political data, they may also have encouraged distorted input designed to influence findings improperly. In support of this view, some of the data from the focus groups were strongly disputed by company management and conflicts of opinion were sometimes evident. Therefore, regardless of these

measures, each input must be considered in a critical, suspicious manner. Preparatory interviews to determine the practices implemented by each team were conducted with project managers, who were also responsible for deploying the agile method in their projects. This may have encouraged depiction of a more comprehensive adoption of practices than might be elicited from other team members. In some interviews, researchers noticed a tendency to 'persuade' the interviewer of the high level of adoption rather than merely describing the method implemented. Finally, because of discussions within the groups before written input was developed, participants are likely to have been influenced in their opinions by other participants – this may have further created distortions and underlines the need for a highly critical appraisal of data by the researcher.

RECOMMENDATIONS

From the experience of conducting the focus groups described earlier and from critical reflection of the researchers after using the seven K&M principles, a set of key recommendations is made to increase the quality of research applying the focus group approach (see Table 2).

CONCLUSION

IS studies using focus group approach often involve many complexities, uncertainties and restrictions. Researchers need to customise and extend focus group approach to suit their specific needs. One of the contributions of this paper is a comprehensive set of recommendations for effectively conducting focus groups where complexity of the concrete situation and participant's context, along with severe time constraints, make data collection difficult. Together with the innovative focus group techniques described, these recommendations will help other researchers to conduct high-quality focus group studies in similar environments. The research project chosen for study was suitable in that it exhibited many of the traits that often render IS research complex and pressurised.

The second contribution of our study is the operationalisation of the Klein & Myers (1999) principles. They have been developed as a more suitable mechanism for the evaluation of interpretive research. Prior to this study however, the principles had not been operationalised, thus limiting their application. The first significant contribution of this study is a set of self-reflective questions operationalising the seven K&M principles. These provide interpretive researchers with practical tools with which to critically reflect, evaluate and consequently improve the quality of their research. While self-reflective questions derived from the positivist concepts of validity and reliability have been widely used for this purpose, as far as the researchers are aware, the questions developed here are the first based on principles specifically proposed in the literature for critical interpretive research. However, we do not claim that the set of self-reflective questions we developed is the only possible way to operationalise

Table 2. Key recommendations for improving the quality of focus group research

Recommendations for effective conduction of focus groups	Recommendation No.	Underlying K&M Principles
Define and agree with participants and researchers the various parts and whole to be considered in the focus group, and structure the focus groups accordingly.	R1, R30	Hermeneutic circle, dialogical reasoning
Perform iterations of data collection and analysis, each informed by the preceding iteration and each validated with participants and other researchers, to ensure that focus group sessions (expensive in terms of ISD team time) are highly contextualised and efficient.	R2, R3, R16, R26, R27, R29, R31	Hermeneutic circle, interaction, dialogical reasoning
In the focus groups, elicit participant input and explicitly describe both the parts themselves (e.g. practices), and how the parts relate to each other as well as to the whole.	R4, R5, R30	Hermeneutic circle, dialogical reasoning
Before the focus group, rigorously establish a comprehensive understanding of context shared across researchers and participants.	R6, R7, R23, R25	Contextualisation, abstraction and generalisation
Where possible, triangulate focus group data with dyadic interviews, direct observation and document analysis. Furthermore, use secondary interviews and peer review by researchers to validate interpretations gained in the focus groups.	R8, R28	Contextualisation, dialogical reasoning
Be aware of the constantly evolving project, team, method and organisational contexts and concrete situation in ISD and explicitly reflect and address how this may affect the focus group research.	R9	Contextualisation
Reflect the ISD team context influencing the focus group session in the research – for example a description of the practices used by the team.	R10	Contextualisation
Given the importance of context in influencing the focus groups, carefully reflect how participants and researchers roles, knowledge and efforts have contributed to the research context, and consider how such 'stakeholding' may affect findings.	R11, R12	Contextualisation
Be cognisant of the influence the perceived research agenda can have on participants, and the effects of participants roles, responsibilities, interests and past efforts may have on the focus groups.	R13, R14	Interaction
Both collection and interpretation of data by more than one researcher and subject to peer review should be used to reduce researcher and participant bias. Researchers should define and play distinct roles in conducting the focus groups, such as facilitation, time keeping and recording discussions.	R15	Interaction
Highly structured focus groups help gather data efficiently but should incorporate some flexibility to accommodate unforeseen topics and complexity. 'Offline' communication between participants and researchers should be accommodated for sensitive inputs.	R17	Interaction

Table 2. cont.

Recommendations for effective conduction of focus groups	Recommendation No.	Underlying K&M Principles
The format of the focus groups should be guided by the theories underlying the research.	R18	Abstraction and generalisation
All researchers and participants should be sensitised to the theoretical concepts before focus group data collection begins.	R19, R20	Abstraction and generalisation
Develop efficient and flexible mechanisms to allow participants associate input with multiple concepts and categories. These may require augmenting the physical placement of post-its on wall posters, such as writing-related concepts directly onto post-its.	R21	Abstraction and generalisation
Participants should be encouraged to illustrate and justify their input by providing 'low inference descriptors' – periodical reminders throughout the focus group, calling out good examples and even marking out special areas of the post-its may help here. To conserve time, participants could add a reminder to a post-it and relate the detail to researchers after the focus group.	R22	Abstraction and generalisation
Reduce variability in context of the case to enhance generalisability across cases.	R24	Abstraction and generalisation
Ensure data collected can be attributed to individuals while maintaining substantial anonymity of written input. For example a code written on the back of each post-it before the focus group could be later used by researchers to identify the contributor.	R25, R34	Abstraction and generalisation, multiple interpretations
Encourage multiple viewpoints of the same events and confront the resulting conflicts and contradictions with participants, perhaps reading back post-its to the group or in follow-up interviews.	R32, R33	Multiple interpretations
Approach data critically – encourage contributions from all participants and look for justifications for opinions. Consider various possible motives, including ulterior, for participants viewpoints – e.g. political correctness or dominant management perspectives – a focus group setting is not conducive to 'teasing out' such issues and follow-up interviews may be necessary.	R35	Suspicion

ISD, information systems development; K&M, Klein & Myers.

the seven K&M principles. On the contrary, we encourage interpretive researchers to adapt or extend our framework, or even to develop new sets of questions, to suit the specific research context and evaluation need of their research. The principles as operationalised here were then demonstrated through the evaluation of a focus group approach used in a large multinational to study agile systems development.

The authors propose further research on both contributions described earlier. The set of self-reflective questions developed in the paper requires further validation by applying them to

other interpretive studies in IS. Additionally, comparing results of evaluations using the K&M-based questions with those based on traditional positivist concepts would identify potential strengths and weaknesses of both approaches, perhaps leading to improvements in both. Also, extending the focus of the evaluation from data collection to other aspects of the research method such as analysis and reporting may be fruitful. In addition, the set of appraisal questions would be further consolidated if other potential evaluation criteria proposed in the IS literature were examined in greater depth, including those from the positivist tradition. These lines of enquiry may prove productive for researchers who are interested in the interpretive research paradigm and research methodology. With regard to the second contribution, the recommendations for conducting focus group studies, these were developed and refined based on three cases of empirical research, specifically with ISD teams and their software development method. Focus groups applied in other IS settings, perhaps with different research objectives, may further extend, strengthen and refine the recommendations.

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